

to provide a holistic view of assets and environments.

PTC ThingWorx. This industrial IoT platform supports the development of digital twins and the integration of real-time data from sensors and devices. It can be used to create digital twins that can interact with the metaverse and provide real-time insights.

What the future holds

Digital twins and the metaverse are two emerging technologies that are expected to shape the future of

city could incorporate live data on traffic, energy usage, and weather conditions.

Personalised virtual spaces.

With the metaverse, individuals can have their own customisable virtual spaces. These spaces can be connected to their digital twins, allowing them to personalise their virtual environment based on their preferences, needs, and interests. Users can also interact with their digital twins within these spaces, creating a personalised and immersive experience.

ences. The metaverse can serve as a platform for virtual commerce, where users can engage in virtual shopping experiences, try out products before purchasing, and interact with virtual representations of real-world stores. Digital twins can enhance these experiences by providing accurate and realistic representations of products and environments.

Cross-platform interoperability. As the metaverse evolves, interoperability between different platforms and systems will become crucial. Digital twins can act as a bridge between various virtual worlds and environments, enabling seamless transfer of data and experiences across different platforms and applications.

Digital twin economies. The metaverse can give rise to new economic models centred around digital twins. Users could create, own, and trade digital twins representing physical assets, virtual goods, or even intellectual property. This could lead to the emergence of digital twin marketplaces and economies, where individuals and businesses can monetise their digital assets.

These trends demonstrate the potential of combining digital twins with the metaverse to create immersive, interconnected, and intelligent digital experiences. However, it's important to note that these trends are speculative, and the actual implementation and adoption of these technologies may vary as they continue to develop.

In conclusion, the metaverse with digital twins has the potential to revolutionise different industrial sectors, but we need to keep in mind certain roadblocks like infrastructure cost, security challenges, compliance issues, and so on. **EFY**

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our digital world. When combined, they can bring about significant advancements and transformative experiences. Here are some potential future trends.

Enhanced virtual collaboration.

Digital twins in the metaverse can enable immersive and collaborative experiences. Users will be able to interact with each other and their digital representations in a shared virtual environment. This can revolutionise remote collaboration, allowing teams from around the world to work together seamlessly.

Real-time data integration. Digital twins can collect real-time data from physical objects or systems and integrate it into the metaverse. This will provide a dynamic and up-to-date representation of the physical world, enabling real-time monitoring, analysis, and simulations. For example, a digital twin of a smart

IoT integration. Internet of Things (IoT) devices can be connected to digital twins within the metaverse, enabling enhanced functionality and control. For instance, a smart home digital twin could integrate with IoT devices such as thermostats, cameras, and appliances, allowing users to remotely monitor and control their physical environment from the metaverse.

AI-driven simulations and predictions. The combination of digital twins and the metaverse can facilitate advanced simulations and predictions powered by artificial intelligence. By leveraging historical data and machine learning algorithms, digital twins can simulate scenarios, predict outcomes, and optimise performance in various domains such as manufacturing, healthcare, and urban planning.

Virtual commerce and experi-